AI & ENGINEERING CAREER MASTER CONTEXT FOR GPT CODEX

OCAI Engineering | Knowledge transfer and project handoff | Version 1.0 | 30 October 2026

Purpose: provide GPT Codex in VS Code with a coherent, reusable understanding of the professional background, learning journey, portfolio, project standards, and long-term objectives discussed with ChatGPT. This is a curated synthesis of available conversation history, not a verbatim archive.

1. Executive profile and destination

• Mechanical engineer practicing since approximately 2015, with hands-on work in mechanical systems, technical inspections, failure analysis, testing, commissioning, process improvement, documentation, and regulated compliance.

• Career transition: Mechanical Engineering -> Data Engineering / Data Science -> applied ML and MLOps -> AI/ML Engineering, with particular interest in RAG, LLM applications, intelligent automation and agentic AI.

• Primary objective: secure international remote work and ultimately relocate to Spain under an appropriate legal route; the family relocation plan is a personal planning objective, not a claim of visa eligibility.

• Professional differentiation: combine domain-specific engineering expertise, regulatory rigor, traceable evidence, quality processes, and applied AI rather than presenting as a generic entry-level coder.

• Work and portfolio identity: OCAI Engineering (Oliveira Costa AI Engineering), bilingual English/Portuguese, oriented toward AI/data/automation and engineering applications.

2. Professional foundation

• Mechanical engineering background includes vehicle inspection, GNV, modifications, periodic inspections, damaged-vehicle processes, technical reports, safety and compliance.

• Technical engineer at Focar, an INMETRO-accredited vehicle inspection organization; holds a small ownership interest but does not have general administrative authority. Do not infer broad permissions from ownership.

• Applied Lean Six Sigma / Green Belt experience: used SIPOC and process improvement to reduce peak-day service time from approximately 60 to 45 minutes (25% reduction).

• Engineering services through OCENGIA include inspection of mechanical equipment, lifting/accessory systems, standards mapping, traceable certificates and reports, and technical approvals.

• Earlier education includes an Executive MBA in Data Science and Big Data (reported completed 2022); current studies include TripleTen MBA in Artificial Intelligence & Data Science.

3. Learning path and present proficiency

• Currently developing practical fluency in Python, pandas, NumPy, scikit-learn, exploratory data analysis, classification, evaluation metrics, preprocessing, encoding, notebooks, environments, and reproducibility.

• Topics explicitly discussed: pandas.get_dummies(), dummy-variable trap, confusion_matrix(), ML classification, feature preparation, Python project organization, RAG fundamentals, LlamaIndex vs LangChain.

• Current working tools: VS Code, Anaconda / Python environments, Git/GitHub, Jupyter-style notebooks, n8n on local Windows, ChatGPT and GPT Codex.

• Future competencies to build, not assume mastered: SQL/data modeling, Docker, cloud (e.g. AWS), orchestration, dbt/Airflow, distributed data tools where relevant, CI/CD, ML experiment tracking, deployment, MLOps, observability, vector databases, RAG evaluation, agent security.

• Learning style: explain underlying reasoning and trade-offs, build real implementations incrementally, verify results, and preserve learning value rather than blindly generating complete solutions.

• English is the preferred language for this career track. Correct unnatural English briefly and offer a more idiomatic professional alternative without derailing technical tasks.

4. MBA and educational portfolio

• TripleTen MBA projects are being rebuilt or maintained as professional repositories, rather than left only as coursework artifacts.

• Megaline Mobile Plan Classification: supervised classification project; repository megaline-mobile-plan-classification.

• Megaline Plan Revenue Analysis: analysis of mobile plans and revenue; repository megaline-plan-revenue-analysis.

• Additional educational repositories reported: vehicle_sales_analysis, game-sales-strategy-eda, streamlit_project.

• Use the original course instructions, data dictionaries, notebooks and datasets as the source of truth for each project; this PDF does not contain full assignment specifications or results.

• Each project should demonstrate problem definition, data understanding, data validation, exploratory analysis, preprocessing, baseline and improved modeling where relevant, appropriate evaluation, limitations, reproducibility and a concise business interpretation.

5. Current applied AI portfolio

MIV - Manual da Inspecao Veicular: domain-grounded vehicle-inspection knowledge assistant and future RAG system. Repository: vehicle-inspection-knowledge-assistant. Operational workflow spans service order intake, document verification, inspection form, photographic evidence, and inspection certificates. Sources include applicable CONTRAN, INMETRO, ABNT and internal technical instructions. Preserve normative version, clause/source, effective date, and engineer validation. One reported MVP_GNV worksheet contains 12 requirements pending technical validation; do not present as approved.

Agent Mike Ross: separate automation/agent project for Focar workflows. Consumes MIV as reference knowledge, but is not the same product. Workflows discussed include calibration register FORM-ADM-017 and monthly FORM-ADM-030. Any email or document delivery requires explicit workflow authorization, source validation and confirmation of actual tool execution.

ATLAS - by OCAI Engineering: mechanical equipment inspection and standards/evidence assistant, explicitly excluding automotive inspections (MIV scope). Workflow: manuals -> current applicable standards -> inspection forms -> measurements, spreadsheets and photographs -> human technical review -> inspection certificate / integrative report. Repository: https://github.com/pedrolocosta/atlas-ocaiengineering . RAG, retrieval evaluation, and agent automation are development goals, not established production capabilities.

Future camera-based operational analytics: possible Focar computer-vision project for inspection workflow timing, bottlenecks and Lean Six Sigma KPIs. Requires formal workplace permissions, privacy/labor compliance, careful retention rules, and human oversight before any camera/data integration.

Paused project: WhatsApp automation with n8n. Do not prioritize over MBA, MIV, Mike Ross, and ATLAS without an explicit change of plan.

6. OCAI brand, public portfolio and boundaries

• Brand: OCAI Engineering, a personal AI/data/automation professional portfolio. Website: https://ocaiengineering.com ; domain ocaiengineering.com; Portuguese domain ocaiengenharia.com.br.

• Website repository: OCAI-website; English and Portuguese pages; Projects & Knowledge should have strong visibility, alongside Engineering Services. External links should open in a new tab.

• OCENGIA is a distinct engineering-services/commercial identity for existing engineering work, including the Auren engagement. Do not collapse the legal/commercial and personal AI identities into one without confirmation.

• GitHub profile: https://github.com/pedrolocosta ; LinkedIn profile: https://www.linkedin.com/in/pedrolocosta/ . Verify live links before publishing.

• LinkedIn content should communicate authentic engineering-to-AI progression, technical substance, project outcomes, limitations, traceable evidence, and useful visual explanations. Avoid claims that prototypes are deployed or validated unless verified.

7. International career and Spain objective

• Target opportunities: international remote roles in Data Engineering, applied ML, AI/ML Engineering, LLM applications, AI automation and agentic systems; choose credible entry roles based on demonstrable skills.

• Medium-term relocation ambition: Spain, with a digital-nomad-related route under consideration. Previously discussed personal target around mid-2027 and staged family move; dates remain tentative.

• Cities considered for scouting: Alicante, Valencia, Zaragoza and Santander; priorities include moderate climate, family life, schools, safety, healthcare, cost and European travel connections.

• Professional public messaging should emphasize availability for international/remote work; do not unnecessarily disclose private family/visa details on LinkedIn.

• Career strategy: create credible public artifacts, improve technical English, develop practical cloud/deployment skills, build an evidence-based CV and apply to suitable remote roles.

8. New division of labor: Codex and ChatGPT

Codex inside VS Code - primary development workspace: work directly on MBA project folders; inspect supplied source materials; structure code and notebooks; build project environment; write and maintain README.md, requirements.txt or pyproject.toml, .gitignore, tests and scripts; run validation locally when tools are available; prepare clean commits and explain changes. Never claim a command, test, push or deployment ran unless it actually did.

ChatGPT - strategic and publication workspace: after a project milestone is completed, review exported code, README, metrics, charts and conclusions; help craft LinkedIn post and image; update OCAI website project descriptions and repository links using available authorized tools. Do not invent model scores or business outcomes.

User - owner and reviewer: provides original coursework/data, creates or approves project/repository, reviews technical interpretations, decides scope, authorizes publication and supplies finished files back to ChatGPT.

9. Standard workflow for each MBA repository

• 1. Read the assignment, datasets and existing notebooks; write a short project brief with objective, deliverables, assumptions and acceptance criteria.

• 2. Inspect existing repo and Git status before modifying files; preserve original work and dataset licensing.

• 3. Agree on directory structure; separate data/raw, data/processed (as appropriate), notebooks, src, tests, reports and figures. Avoid overengineering small assignments.

• 4. Configure Python environment and dependency pinning; add useful .gitignore patterns for environments, secrets, caches, large/private datasets and notebook checkpoints.

• 5. Reproduce the original analysis, identify leakage and evaluation risks, use train/test separation correctly, document random seeds and validate data transformations.

• 6. Add clear README: context, objective, dataset provenance, methods, how to run, results with real numbers, limitations, future improvements and links.

• 7. Run checks/tests; inspect outputs, code style, notebooks, charts and Git diff; confirm no secrets or unauthorized/private datasets are committed.

• 8. User reviews and commits/pushes with Codex support; then export a publication handoff bundle to ChatGPT.

10. Publication handoff package to ChatGPT

• Repository URL and commit hash or release tag.

• Final README, requirements/pyproject, .gitignore, representative notebook(s) or Python files.

• Project objective, dataset provenance and permission to share, methods, actual quantitative results, charts, caveats, and key lessons.

• Two or three accurate technical takeaways, intended LinkedIn audience, preferred visual style, and website destination/page.

• Confirm whether the project is coursework, independent extension, prototype or production system.

11. Operational instructions for GPT Codex

You are an AI coding collaborator supporting a mechanical engineer transitioning to AI Engineering. Prioritize teaching, correctness, reproducibility, and truthful documentation. Start by inspecting the actual repository and task materials. Ask for missing critical information; do not fabricate datasets, model results, files, commits or tests. Explain technical decisions and alternatives briefly. Implement in manageable increments; preserve the user's original work and request approval for destructive changes. Respect data privacy, licenses, confidentiality, professional standards and engineering sign-off. Keep code and docs in professional English; offer brief language corrections when appropriate. Distinguish existing functionality, prototypes, plans, and validated results. The user will publish only after review.

12. Current priorities and near-term roadmap

• Immediate: transition TripleTen MBA projects to VS Code + Codex with clean repositories, reproducible execution and documented learning.

• Parallel: continue MIV, Mike Ross and ATLAS according to available time and project needs, avoiding unsupported production claims.

• Learning progression: stronger Python/pandas/scikit-learn -> data engineering fundamentals -> robust RAG with LlamaIndex -> LangChain when justified -> deployment, evaluation, observability and MLOps.

• Public output: regular but substantive GitHub/LinkedIn updates through OCAI Engineering, based on completed verified work.

• Career outcomes: remote international applications supported by measurable project artifacts and a coherent engineering-to-AI narrative.

13. Knowledge limits and update protocol

This briefing synthesizes information available to ChatGPT as of 30 October 2026, including prior discussions and recalled project context. It is not an exhaustive export of every chat, attachment, codebase, OneNote page, Drive file or repository. Details such as current implementation status, GitHub branches, source datasets, legal requirements, external website content and project results must be rechecked against primary files and authoritative sources. This document may contain historical goals or tentative timelines. Treat user-supplied current repository files and explicit new decisions as more authoritative.

Recommended update practice: maintain a small CONTEXT.md or PROJECT_BRIEF.md per repository, with last-updated date, decisions, current state, known issues and next action. Refresh this master briefing when career strategy or portfolio priorities change.

14. Selected repository and website references

https://github.com/pedrolocosta

https://github.com/pedrolocosta/vehicle-inspection-knowledge-assistant

https://github.com/pedrolocosta/atlas-ocaiengineering

https://github.com/pedrolocosta/megaline-mobile-plan-classification

https://github.com/pedrolocosta/megaline-plan-revenue-analysis

https://github.com/pedrolocosta/OCAI-website

https://ocaiengineering.com